

Smart Document Q&A System with Agentic RAG

[Host the application in front end of your choice and attach the response Screenshots along with your

. ipynbfiles – file naming (**empid_Name.ipynb**)]

PROBLEM STATEMENT

Build an intelligent document analysis system that can automatically ingest a collection of documents given and build a searchable knowledge base and answer complex questions by retrieving and synthesizing information from multiple sources.

BUSINESS CONTEXT

Imagine you're building a system for a research lab where scientists need to quickly find information across hundreds of research papers, or for a legal firm that needs to search through thousands of case documents. Users should be able to ask complex questions and get accurate answers with proper source citations.

CORE REQUIREMENTS

1. True Agentic RAG Implementation

- **Document Ingestion Pipeline:** Automatically process uploaded PDF/text files and break them into semantic chunks
- **Vector Database:** Store document embeddings using ChromaDB (free) for semantic similarity search
- **Smart Retrieval:** Use embedding similarity to find relevant chunks across multiple documents
- **Context Assembly:** Combine multiple retrieved chunks to provide comprehensive answers
- **Source Attribution:** Always cite which documents and sections provided the information
- **Iterative Refinement:** Agent should ask follow-up questions or search for additional context when initial retrieval is insufficient

WORKFLOW

The agent should:

Ingestion Phase

- Accept document uploads (PDFs, text files)
- Extract and clean text content
- Create semantic chunks with overlap
- Generate embeddings and store in vector database

Query Phase

- Parse user questions and create query embeddings
- Retrieve top-k most relevant chunks across documents
- Assess if retrieved context is sufficient to answer the question
- If insufficient, reformulate search or ask clarifying questions

Generation Phase

- Synthesize answer using retrieved chunks as context
- Provide source citations with document names and page numbers
- Identify confidence level and knowledge gaps

How it will be evaluated: (40 Points)

- 1) **Framework and Tech stack usage** – 10 points
- 2) **RAG architecture and implementation** – 15 points
- 3) **Additional refinements for enhanced outputs** – 10 Points (Prompt rewriting, Advanced prompting techniques, reranking response, guardrails for output response)
- 4) **Front end** – 5 Points

Pdf Documents to be used: Attached in the folder

Questions: (Any 2 question's response is required)

- 1) What was the total revenue reported by the Roy J. Carver Charitable Trust for FY2022?
- 2) How much did the Trust report in total expenses during FY2022?
- 3) What is the net change in assets for FY2022 compared to the prior year?
- 4) What types of grants or program services did the Trust highlight in the notes to the financial statements?
- 5) What auditing firm provided the independent auditor's report for FY2022?

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